SOFTWARE ENGINEERING LAB 1

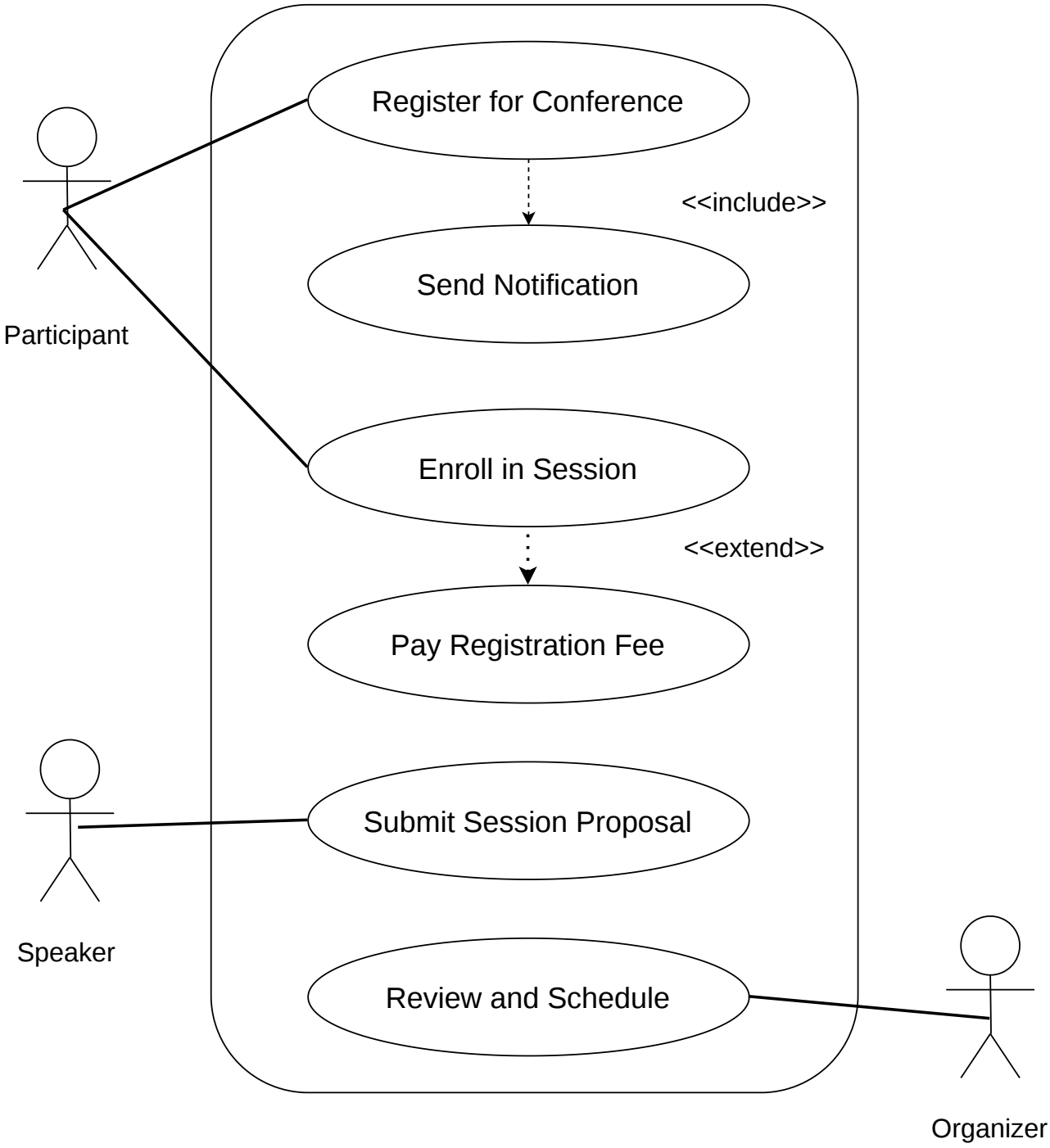
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SECTION G

PROBLEM STATEMENT 51
Virtual Technical Conference Platform

1. Requirements Table :

REQ ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE
FR-001	Functional	The system shall allow a participant to register for the conference by providing personal details and creating an account.	High	Pass: successful registration creates an account and sends a confirmation notification. Fail: duplicate email is accepted without a validation error.	Registration is the entry point for all participant activity.
FR-002	Functional	The system shall allow registered participants to browse the list of technical sessions and enroll in sessions of their choice.	High	Pass: an enrolled session appears in the participant's schedule. Fail: a participant can enroll in two sessions with overlapping time slots without a warning.	Core browsing and enrollment functionality described in the scenario.
FR-003	Functional	The system shall require participants to pay the registration fee through the payment gateway before granting access to paid sessions.	High	Pass: a paid session cannot be joined without a confirmed payment status. Fail: a session is accessible while payment is pending or failed.	Directly required by the scenario for paid sessions.
FR-004	Functional	The system shall allow speakers to create a profile, submit session proposals, and upload presentation materials for review.	Medium	Pass: a submitted proposal appears in the organizer's review queue linked to the speaker profile. Fail: a proposal is submitted without an associated speaker profile.	Supports the speaker side of the workflow.

FR-005	Functional	The system shall allow the conference organizer to review submitted proposals, accept or reject them, assign accepted speakers to time slots, and publish the final schedule.	High	Pass: the published schedule shows all accepted proposals with assigned time slots and no double booking. Fail: two speakers are assigned to the same slot without a conflict alert.	Organizer scheduling is required to produce a usable conference agenda.
NFR-001	Nonfunctional, Performance	The live session chat facility shall deliver participant messages to all attendees within 2 seconds under a load of 500 concurrent users per session.	High	Pass: load testing confirms delivery latency under 2 seconds at 500 concurrent users. Fail: latency exceeds 2 seconds or messages are dropped under load.	Real time interaction depends on low chat latency during live sessions.
NFR-002	Nonfunctional, Security	All payment transactions and participant personal data shall be encrypted in transit and at rest.	High	Pass: a security review confirms TLS in transit and AES-256 at rest with no plaintext personal data in logs. Fail: any personal data or payment data is found unencrypted in storage or transmission.	Protects sensitive participant and payment data throughout the platform.



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3. Use-Case Flow Document :

Use Case: Enroll in Session

Actors: Participant, Payment Gateway

Preconditions: The participant has registered and is logged in. The conference schedule has been published.

Postconditions: The participant is enrolled in the session. If the session is paid, the payment is confirmed. The participant receives a confirmation notification.

Main Success Scenario:

1. Participant browses the list of published technical sessions.
2. Participant selects a session and requests to enroll.
3. System checks whether the session requires a paid registration fee.
4. If the session is free, system enrolls the participant directly and skips to step 9.
5. If the session is paid, system displays a payment form requesting amount and payment method.
6. Participant selects a payment method and submits payment details.
7. System connects to the Payment Gateway.
8. Payment Gateway authorizes the payment.
9. System confirms enrollment and adds the session to the participant's schedule.
10. System sends a registration confirmation notification to the participant.
11. Use case ends successfully.

Alternate Flow:
8a. Payment Declined
8a1. System displays a payment failed message.
8a2. Participant is prompted to try another payment method.
8a3. If payment fails again after two attempts, the enrollment is cancelled and the participant is notified.